

Predicting Donor Behavior: Applying Machine Learning to Donor Classification

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Current State of Campaign Finance²

- ▶ Campaign Expenditures are rising:
 - ▶ 2000: \$5 Billion.
 - ▶ 2020: \$15 Billion.
- ▶ Campaign Expenditures correlate with electoral success.
- ▶ Individual Donors are a primary source of funding.
 - ▶ Provided about 60% of direct funding in 2020.

²Glavin (2024); Reynolds and Green (2023); Stratmann (2009); Gordon and Hartmann (2013); Baker (2022); OpenSecrets.org (2020); Dwyre and Kolodny (2024)

Role for Predictive Models³

- ▶ Donors are scarce:
 - ▶ 8.5% donated in 2020.
 - ▶ Less than 2% gave more than \$200.
- ▶ Predictive models save campaign time and money.
- ▶ Prominent usage since 2008 but little academic research.

³Dwyre and Kolodny (2024); Walker and Nowlin (2021); Nickerson and Rogers (2014)

Building Predictive Models

- ▶ Classification requires informative features
- ▶ Tested 3 difference models:
 - ▶ Logistic regression.
 - ▶ Random forest.
 - ▶ Gradient boosted trees.
- ▶ Main Questions:
 - ▶ Will these models be effective?
 - ▶ Which model will perform the best?
 - ▶ Which features are the most informative?

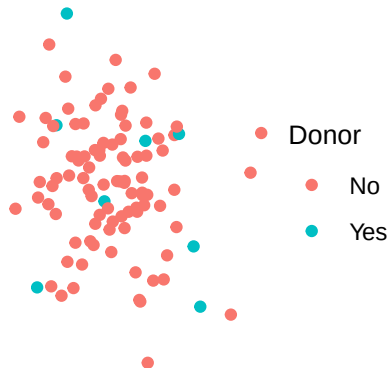


Figure 1: Example Classification Scenarios

Donor Motivations⁴

- ▶ Theorists have split donors into categories:
 - ▶ Investors.
 - ▶ Ideologues.
 - ▶ Intimates.
- ▶ Key empirical features:
 - ▶ Ideological fervor (Candidates and Voters).
 - ▶ Policy and partisan alignment.
 - ▶ Connections to other donors.

⁴Dwyre and Kolodny (2024); Francia et al. (2003); Baker (2022); M. J. Barber, Canes-Wrone, and Thrower (2017); M. Barber (2016); Niebler and Urban (2017); Ensley (2009); Traag (2016)

Data Description⁵

- ▶ Data from 2006-2008, 2012, 2014, 2016-2020, and 2022 CES surveys.
 - ▶ $n = 527156$
 - ▶ 26.18% are donors.
- ▶ 27 features:
 - ▶ Categorical: 138 one-hot-coded.
 - ▶ Demographics.
 - ▶ Wealth.
 - ▶ Political Engagement.
 - ▶ Ideology.
- ▶ Training-Test Split
 - ▶ 80% training.
 - ▶ 20% test.

⁵Ansolahehere (2010); Ansolahehere (2011); Ansolahehere (2013); Ansolahehere (2012); Ansolahehere and Schaffner (2013); Schaffner and Ansolahehere (2015); Ansolahehere and Schaffner (2017); Schaffner and Ansolahehere (2019); Schaffner, Ansolahehere, and Luks (2019); Ansolahehere, Schaffner, and Luks (2020); Schaffner, Ansolahehere, and Luks (2021); Schaffner, Ansolahehere, and Shih (2023); Lones (2024)

Hypotheses

- ▶ H_1 : At least one of the models will perform successfully.
- ▶ H_2 : The more complex models will perform better relative to less complex models.
 - ▶ Donor behavior is complex.
- ▶ H_3 : Political and civic engagement variables will have high predictive power:
 - ▶ Recognizing elected officials.
 - ▶ Political activities.
 - ▶ Voting intent.

Missing Data⁶

- ▶ Missing outcomes dropped
 - ▶ $n = 451480$
- ▶ Missingness mechanism assumption: Missing at Random (MAR)
- ▶ Multiple Imputation By Chained Equations (MICE):
 - ▶ 50 iterations.
 - ▶ Train and test imputed separately.
 - ▶ mice package in R.

Table 1: Glimpse of Missing Variables

	Variable	Number
1	News Activites	80,782
2	Contacted	76,588
3	Donated	75,502
4	Political Activities	69,278
5	Income	69,021
6	Education	74
7	Race	5
8	Gender	0
9	Union	0
10	Age	0

⁶Shaheen MZ. Memon, Wamala, and Kabano (2023); Shaheen M. Z. Memon, Wamala, and Kabano (2022); Zhang (2022); Vink et al. (2014); Austin and van Buuren (2023); van Buuren et al. (2024); Lones (2024)

Class Weighting⁷

- ▶ Significant class imbalance
 - ▶ 26% are donors.
 - ▶ 74% are not donors.
- ▶ Sample Based Class Weight (SBCW) method:

$$W_i = \frac{N}{k * n_i}$$

$$W_{\text{donor}} = \frac{361323}{2 * 94548} = 1.9108 \quad W_{\text{non-donor}} = \frac{361323}{2 * 266775} = 0.6772$$

⁷Kuhn and Johnson (2016); Bakirarar and Elhan (2023)

Models Used⁸

- ▶ Logistic Regression:

- ▶ `glm()` from R

$$\hat{y}_i = \frac{1}{1 + \exp(-\mathbf{x}_i^T \boldsymbol{\beta})}$$

- ▶ Random Forest:

- ▶ `ranger()` from `ranger` package in R.

$$\hat{y}_i = \frac{1}{T} \sum_{j=1}^N f_j(\mathbf{x}_i) = \frac{\text{Tree}_1 + \text{Tree}_2 + \dots + \text{Tree}_T}{T}$$

- ▶ Gradient Boosted Trees:

- ▶ `LightGBM` implementation in R

$$\hat{y}_i = f_0(\mathbf{x}_i) + \sum_{t=1}^T f_t(\mathbf{x}_i) =$$

Initial Guess + Tree₁ + Tree₂ + ... + Tree_T

⁸James et al. (2013a); James et al. (2013b); Breiman (2001); Friedman (2001); Chen and Guestrin (2016); Ke et al. (2017)

Hyperparameter Tuning⁹

- ▶ 10-fold CV on the training set
- ▶ Stage 1. Random hyperparameter search.
- ▶ Stage 2. 50 iterations of Bayesian Optimization.

Table 2: Hyperparameter Ranges Tested

Model	Hyperparameter	Range
Both	Mtry	1 – 27
Both	Trees	100 – 3000
Both	Min Node Size	1 – 1000
LightGBM	Tree Depth	3 – 25
LightGBM	Learn Rate	10^{-10} – 10^{-1}
LightGBM	Loss Reduction	10^{-10} – $10^{1.5}$

⁹Kuhn (n.d.); Arnold et al. (2024); Probst, Wright, and Boulesteix (2019); Raschka (2020); Bergstra and Bengio (2012); Wu et al. (2019)

Test Set Performance

Table 3

	Metric	Logit	Gradient Boosted	Random Forest
1	Accuracy	0.7590	0.7706	0.7868
2	F1	0.6270	0.6383	0.6386
3	Recall	0.7712	0.7707	0.7174
4	Precision	0.5282	0.5448	0.5755
5	ROC-AUC	0.8432	0.8529	0.8517

0.01 corresponds to 900 (Acc.), 237 (Rec.), and 326 (Prec.) People.

Model Comparisons¹⁰

- ▶ $\alpha = 0.05$
- ▶ Cochran's Q Test:
 - ▶ $H_0 : ACC_1 = ACC_2 = ACC_3$
 - ▶ $H_a : \exists i \neq j$ such that $ACC_i \neq ACC_j$, for $i, j \in \{1, 2, 3\}$
 - ▶ $p = 2.18 \times 10^{-196}$
- ▶ Pairwise McNemar:
 - ▶ $H_0 : ACC_i = ACC_j$
 - ▶ $H_a : ACC_i \neq ACC_j$
 - ▶ Logit and boosted trees: 4.97×10^{-37}
 - ▶ Random forest and boosted trees: 9.34×10^{-70}
 - ▶ Logit and random forest: 4.34×10^{-184}

Select Variable Contributions: Permutation Importance

Table 4

	Variable	Logit	Random Forest	Gradient Boosted
1	Political Activities	0.0510	0.0480	0.0490
2	Contacted	0.0066	0.0067	0.0064
3	News Activities	0.0046	0.0017	0.0039
4	Party ID	0.0040	0.0047	-0.0021
5	Income Range	0.0034	0.0031	0.0060
6	Own Home	0.0029	0.0010	0.0024
7	Education	0.0024	0.0030	0.0026
8	Ideology	0.0018	0.0009	-0.0038
9	Vote Intent	0.0013	-0.0008	0.0020
10	Investor	0.0008	0.0016	0.0018
11	Vote Prior	0.0006	-0.0003	-0.0007
12	Race	0.0004	0.0008	0.0032
13	Ran For Office	0.00004	0.0002	0.0007
14	Political Interest	-0.0002	0.0022	-0.0007
15	Voter Registration	-0.0017	-0.0010	-0.0013

Revist Research Questions and Hypotheses

- ▶ Q1: Were these models effective?
 - ▶ Potentially, but without a proper comparison its difficult to conclude anything.
- ▶ Q2: Which model was the best?
 - ▶ F1 and Precision: Random Forest.
 - ▶ Recall and ROC-AUC: Gradient Boosted.
- ▶ Q3: Which features were the most important to predictive power?
 - ▶ Political Activities.
 - ▶ Model Dependent.
 - ▶ Ideology, Party ID, Voting Intent

Limitations

- ▶ Self reporting, missingness, other biases from survey.
- ▶ Limited nuance in “Donor” variable.
 - ▶ Contribution size.
 - ▶ Contribution frequency.
 - ▶ Contribution recipient.
- ▶ Aggregating across election years.
 - ▶ Changing political and campaign finance landscape.

Future Research and Next Steps:

- ▶ More comprehensive data:
 - ▶ Non-free sources.
 - ▶ State or election specific.
 - ▶ Specifically engineered surveys.
- ▶ Utilizing LLMs
 - ▶ LLMs can view the data in the raw form, and have deeper embeddings for qualitative data.
 - ▶ Vast array of Pre-trained models.
 - ▶ Low Rank Adaptation to retrain.

Thank You

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References I

- Ansolabehere, Stephen. 2010. "CCES Common Content, 2006." doi:10.7910/DVN/Q8HC9N.
- Ansolabehere, Stephen. 2011. "CCES Common Content, 2007."
<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/OOXTJ5> (February 13, 2025).
- Ansolabehere, Stephen. 2012. "CCES Common Content, 2010."
<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VKKRWA> (February 13, 2025).
- Ansolabehere, Stephen. 2013. "CCES, Common Content, 2008."
<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/YUYIVB> (February 13, 2025).
- Ansolabehere, Stephen, and Brian Schaffner. 2013. "CCES Common Content, 2012." doi:10.7910/DVN/HQEVPK.
- Ansolabehere, Stephen, and Brian Schaffner. 2017. "CCES Common Content, 2016." doi:10.7910/DVN/GDF6Z0.
- Ansolabehere, Stephen, Brian Schaffner, and Samantha Luks. 2020. "CCES Common Content, 2019."
doi:10.7910/DVN/WOT708.
- Arnold, Christian, Luka Biedebach, Andreas Küpfer, and Marcel Neunhoeffler. 2024. "The Role of Hyperparameters in Machine Learning Models and How to Tune Them." *Political Science Research and Methods* 12(4): 841–48.
doi:10.1017/psrm.2023.61.
- Austin, Peter C, and Stef van Buuren. 2023. "Logistic Regression Vs. Predictive Mean Matching for Imputing Binary Covariates." *Statistical Methods in Medical Research* 32(11): 2172–83. doi:10.1177/09622802231198795.
- Baker, Anne. 2022. "Policies, Profits, Networks, or Duty? Donors' Motivations for Contributing to Parties and Interest Groups." *The Social Science Journal* 59(4): 630–45. doi:10.1080/03623319.2020.1727224.

References II

- Bakirarar, Batuhan, and Atilla Halil Elhan. 2023. "Class Weighting Technique to Deal with Imbalanced Class Problem in Machine Learning: Methodological Research." *Türkiye Klinikleri Biyoistatistik* 15(1): 19–29. doi:10.5336/biostatic.2022-93961.
- Barber, Michael. 2016. "Donation Motivations: Testing Theories of Access and Ideology." *Political Research Quarterly* 69(1): 148–59. doi:10.1177/1065912915624164.
- Barber, Michael J., Brandice Canes-Wrone, and Sharece Thrower. 2017. "Ideologically Sophisticated Donors: Which Candidates Do Individual Contributors Finance?" *American Journal of Political Science* 61(2): 271–88. <https://www.jstor.org/stable/26384731> (February 12, 2025).
- Bergstra, James, and Yoshua Bengio. 2012. "Random Search for Hyper-Parameter Optimization." *Journal of Machine Learning Research* 13(10): 281–305. <http://jmlr.org/papers/v13/bergstra12a.html>.
- Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45(1): 5–32. doi:10.1023/A:1010933404324.
- van Buuren, Stef, Karin Groothuis-Oudshoorn, Gerko Vink, Rianne Schouten, Alexander Robitzsch, Patrick Rockenschaub, Lisa Doove, et al. 2024. "Mice: Multivariate Imputation by Chained Equations." <https://cran.r-project.org/web/packages/mice/index.html> (February 25, 2025).
- Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–94. doi:10.1145/2939672.2939785.
- Dwyre, Diana, and Robin Kolodny. 2024. *The Fundamentals of Campaign Finance in the U.S.: Why We Have the System We Have*. Ann Arbor, MI: University of Michigan Press. doi:10.3998/mpub.9813302.
- Ensley, Michael J. 2009. "Individual Campaign Contributions and Candidate Ideology." *Public Choice* 138(1/2): 221–38. <https://www.jstor.org/stable/40270840> (February 12, 2025).

References III

- Francia, Peter L., Paul S. Herrnson, John C. Green, Lynda W. Powell, and Clyde Wilcox. 2003. *The Financiers of Congressional Elections: Investors, Ideologues, and Intimates*. New York: Columbia University Press.
- Friedman, Jerome H. 2001. "Greedy Function Approximation: A Gradient Boosting Machine." *The Annals of Statistics* 29(5): 1189–1232. doi:10.1214/aos/1013203451.
- Glavin, Sarah Bryner and Brendan. 2024. "Total 2024 Election Spending Projected to Exceed Previous Record." <https://opensecretsnews.wpcomstaging.com/2024/10/total-2024-election-spending-projected-to-exceed-previous-record/> (April 16, 2025).
- Gordon, Brett R., and Wesley R. Hartmann. 2013. "Advertising Effects in Presidential Elections." *Marketing Science* 32(1): 19–35. <https://www.jstor.org/stable/23361432> (April 15, 2025).
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani, eds. 2013a. "Logistic Regression." In *An Introduction to Statistical Learning: With Applications in R*, Springer texts in statistics, New York: Springer, 133–40.
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani, eds. 2013b. "Tree Based Methods." In *An Introduction to Statistical Learning: With Applications in R*, Springer texts in statistics, New York: Springer, 327–403.
- Ke, Guolin, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." In *Advances in Neural Information Processing Systems*, Curran Associates, Inc. https://papers.nips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html (August 4, 2025).
- Kuhn, Max. *The Caret Package*. <https://topepo.github.io/caret/> (March 4, 2025).
- Kuhn, Max, and Kjell Johnson. 2016. *Applied Predictive Modeling*. Corrected at 5th printing. New York: Springer.

References IV

- Lones, Michael A. 2024. "Avoiding Common Machine Learning Pitfalls." *Patterns* 5(10). doi:10.1016/j.patter.2024.101046.
- Memon, Shaheen M. Z., Robert Wamala, and Ignace H. Kabano. 2022. "Missing Data Analysis Using Statistical and Machine Learning Methods in Facility-Based Maternal Health Records." *SN Computer Science* 3(5): 355. doi:10.1007/s42979-022-01249-z.
- Memon, Shaheen MZ., Robert Wamala, and Ignace H. Kabano. 2023. "A Comparison of Imputation Methods for Categorical Data." *Informatics in Medicine Unlocked* 42: 101382. doi:10.1016/j.imu.2023.101382.
- Nickerson, David W., and Todd Rogers. 2014. "Political Campaigns and Big Data." *Journal of Economic Perspectives* 28(2): 51–74. doi:10.1257/jep.28.2.51.
- Niebler, Sarah, and Carly Urban. 2017. "Does Negative Advertising Affect Giving Behavior? Evidence from Campaign Contributions." *Journal of Public Economics* 146: 15–26. doi:10.1016/j.jpubeco.2016.12.001.
- OpenSecrets.org. 2020. "2020 Election to Cost \$14 Billion, Blowing Away Spending Records." <https://www.opensecrets.org/news/2020/10/cost-of-2020-election-14billion-update/> (January 7, 2026).
- Probst, Philipp, Marvin N. Wright, and Anne-Laure Boulesteix. 2019. "Hyperparameters and Tuning Strategies for Random Forest." *WIREs Data Mining and Knowledge Discovery* 9(3): e1301. doi:10.1002/widm.1301.
- Raschka, Sebastian. 2020. "Model Evaluation, Model Selection, and Algorithm Selection in Machine Learning." doi:10.48550/arXiv.1811.12808.
- Reynolds, Molly, and John Clifford Green, eds. 2023. *Financing the 2020 Election*. Washington, DC: Brookings Institution Press.

References V

- Schaffner, Brian, and Stephen Ansolabehere. 2015. "CCES Common Content, 2014." : 107960901, 2009825, 1304010, 486343, 109447071, 248189, 486168, 523682. doi:10.7910/DVN/XFXJVY.
- Schaffner, Brian, Stephen Ansolabehere, and Sam Luks. 2019. "CCES Common Content, 2018." doi:10.7910/DVN/ZSBZ7K.
- Schaffner, Brian, Stephen Ansolabehere, and Sam Luks. 2021. "Cooperative Election Study Common Content, 2020." doi:10.7910/DVN/E9N6PH.
- Schaffner, Brian, Stephen Ansolabehere, and Marissa Shih. 2023. "Cooperative Election Study Common Content, 2022." doi:10.7910/DVN/PR4L8P.
- Schaffner, Brian, and Stephen Ansolabehere. 2019. "2017 CCES Common Content." doi:10.7910/DVN/3STEZY.
- Stratmann, Thomas. 2009. "How Prices Matter in Politics: The Returns to Campaign Advertising." *Public Choice* 140(3): 357–77. doi:10.1007/s11127-009-9425-z.
- Traag, Vincent A. 2016. "Complex Contagion of Campaign Donations." *PLoS ONE* 11(4): e0153539. doi:10.1371/journal.pone.0153539.
- Vink, Gerko, Laurence E. Frank, Jeroen Pannekoek, and Stef van Buuren. 2014. "Predictive Mean Matching Imputation of Semicontinuous Variables." *Statistica Neerlandica* 68(1): 61–90. doi:10.1111/stan.12023.
- Walker, Doug, and Edward L. Nowlin. 2021. "Data-Driven Precision and Selectiveness in Political Campaign Fundraising." *Journal of Political Marketing* 20(2): 73–92. doi:10.1080/15377857.2018.1457590.
- Wu, Jia, Xiu-Yun Chen, Hao Zhang, Li-Dong Xiong, Hang Lei, and Si-Hao Deng. 2019. "Hyperparameter Optimization for Machine Learning Models Based on Bayesian Optimizationb." *Journal of Electronic Science and Technology* 17(1): 26–40. doi:10.11989/JEST.1674-862X.80904120.

References VI

Zhang, Shichao. 2022. "Challenges in KNN Classification." *IEEE Transactions on Knowledge and Data Engineering* 34(10): 4663–75. doi:10.1109/TKDE.2021.3049250.